

ASHISH JOSHI

✉ ashishjoshirj57@gmail.com [in LinkedIn](#) [GitHub](#) [Portfolio](#) ☎ +91-9625497741

Education

Netaji Subhas University of Technology (NSUT)

B.Tech in Computer Science, AI Specialization

CGPA: 8.61/10.0

Aug 2024 – May 2028

Senior Secondary (CBSE)

Kendriya Vidyalaya A.A.I Rangpuri, New Delhi

94.2%

2023

Secondary (CBSE)

Heera Public School, New Delhi

92.3%

2020

Achievements & Activities

Microsoft Devnovate Hackathon

[Certificate](#)

Competed in 24-hour hackathon at Microsoft Office Noida, building full-stack AI applications under time constraints

CS50 SQL - Harvard University

[Certificate](#) | [Project Demo](#)

Completed advanced database course covering design, indexing, query optimization, and transaction management

Competitive Programming: Active on LeetCode and GfG, solved 300+ problems across dynamic programming, graph algorithms, and various difficulty levels

Axiom NSUT – Philosophical Society: Active member engaging in structured intellectual discourse and debates, developed critical thinking, communication, and leadership skills through organizing and participating in society events

Technical Skills

Languages: Python, C++, JavaScript (ES6+), SQL, HTML/CSS, C

Frameworks & Libraries: React.js, Next.js, FastAPI, Node.js, PyTorch, Scikit-learn, NumPy, Pandas

AI/ML: LLMs, RAG, CNNs, NLP, Vector Embeddings

Tools & Databases: Git, Docker, Linux, REST APIs, PostgreSQL, FAISS, Firebase

Cloud & Deployment: Vercel, Railway, Hugging Face

Projects

LectureIQ – Multi-Modal AI Lecture Study Platform | [GitHub](#)

May 2026

FastAPI, React, Supabase, Groq API, LLMs, Vector Databases, Docker

- Architected a full-stack AI platform that processes lecture videos, automating audio transcription to dynamically generate structured markdown notes, interactive quizzes, and personalized study guides
- Engineered an intelligent routing system using various LLM APIs, dynamically allocating lightweight models for text formatting and advanced models for complex reasoning to strictly optimize token usage and prevent API timeouts
- Integrated vector databases to power a highly contextual Retrieval-Augmented Generation (RAG) chat assistant, enabling students to query lecture transcripts and receive instant, grounded answers
- Implemented real-time Server-Sent Events (SSE) to stream processing progress and chunked AI responses directly to the React frontend, ensuring a seamless and highly responsive user experience
- Deployed a scalable containerized backend utilizing FastAPI and Docker, engineered with custom fallback logic to gracefully handle rate limit errors without dropping active user connections

AI-Powered Digital Forensic Investigation Agent | [GitHub](#) | [Demo](#)

Jan 2025

FastAPI, React, Next.js, TypeScript, RAG, FAISS, Microsoft Phi-3

- Engineered an end-to-end AI agent leveraging RAG architecture to analyze Universal Forensic Data Reports (UFDR), enabling investigators to query 10,000+ forensic data points using natural language
- Architected a custom multi-format parser to extract and structure data from XML and ZIP-based UFDR files, reducing manual data processing time by 85% for law enforcement workflows
- Integrated Microsoft Phi-3 Small Language Model with FAISS vector database for semantic search, achieving sub-second query response times across large datasets
- Developed full-stack application with FastAPI backend and Next.js/TypeScript frontend, implementing real-time streaming responses and automated evidence report generation
- Deployed scalable microservices on Railway (backend) and Vercel (frontend) with optimized API endpoints serving concurrent requests